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Uncertainty-aware evidential learning for legal case retrieval with noisy correspondence ☆

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ABSTRACT

Legal case retrieval is a critical task in intelligent legal systems, providing relevant precedents to assist judges in their decision-making. While current data-driven neural retrieval methods have demonstrated impressive performance on clean, annotated data, they often ignore the robustness against noisy correspondences. In practice, legal annotators are required to identify *legal uncertainty*, which refers to the ambiguity or unpredictability in legal interpretations and applications, in relevance estimation between cases. This uncertainty often introduces noise into the training data, leading to unreliable predictions and potentially impacting the fairness and justice of downstream tasks. Focusing on this robustness issue, we propose a novel evidential learning framework called ELCR, which explicitly models the *legal uncertainty* and addresses noisy correspondences. Specifically, we first estimate the multi-faceted relevance between query-candidate cases from the concept, rule, and fact levels. These relevance estimations are then used to obtain the evidence-based uncertainty under the Dempster-Shafer Evidence Theory, which helps correct labels from noisy correspondence. Guided by two elaborate evidence-based training objectives, ELCR provides accurate uncertainty estimation, enhancing reliability and robustness. Extensive experiments on various noise proportions across two benchmark datasets demonstrate that our method exhibits robustness against noisy correspondences while maintaining competitive retrieval performance.

1. Introduction

Legal Case Retrieval (LCR) is a specialized task in the field of information retrieval that focuses on identifying and retrieving legal cases relevant to a particular query case. This task is crucial for legal professionals who need to find precedents and related cases to support their arguments in litigation or legal research.

In recent years, neural retrieval methods have demonstrated their effectiveness in the LCR task. One prominent line of work focuses on introducing external knowledge through network-based methods, such as the Precedent Citation Network (PCNet) [1] and Hier-SPCNet [2], which leverage precedent citation relations to provide relevance signals. Another line of work has concentrated on

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Concept-level example:
Case A: XX had an argument with YY … YY picked up a knife and threatened to harm XX… XX grabbed the knife and accidentally stabbed YY… XX claims that her actions were in justifiable defense.
Case B: YY threw a vase at XX… To dodge, XX accidentally pushed YY over, causing her injury… XX claims that her actions were in justifiable defense.
Judicial interpretation:
For Case A, the court ruled that XX's actions constituted justifiable defense because she was protecting herself.
For Case B, the court ruled that XX's actions exceeded the scope of justifiable defense because her reaction was too extreme and did not meet the standards of reasonableness for self-defense.

Rule-level uncertainty example :
Case A: XX led several relatives… attempted to prevent the court from continuing the trial…
Case B: XX threatened a witness, claiming that…
Judicial interpretation:
Case A applies to the Crime of disrupting the court order, Rule 1, Article 309 stipulates the situation of gathering crowds to make noise or assault the court.
Case B applies to the Crime of disrupting the court order, Rule 2, Article 309 stipulates the situation of threatening litigation participants and seriously disrupting court order.

Fact-level uncertainty example :
Case A: … had a dispute… XX, armed with a knife... stabbed towards YY... causing YY to die on the spot. (Intentional homicide)
Case B: … in a verbal altercation… In a state of high emotion, XX picked up a fruit knife from the table and stabbed … YY succumbed to excessive blood loss. (Intentional injury resulting in death)
Judicial interpretation:
For Case A, the court ruled that XX's actions constituted intentional homicide because XX deliberately armed himself with a knife and carried out the stabbing with the intent to kill, as evidenced by the premeditation and lethal nature of the attack.
For Case B, the court ruled that XX's actions constituted intentional injury resulting in death because the stabbing occurred during a moment of high emotion without premeditation, and the primary intent was to cause injury rather than death, resulting in YY's death due to blood loss.

Fig. 1. The illustrations of *legal uncertainty* and the corresponding judicial interpretations from the concept, rule, and fact levels. The red parts indicate uncertain correspondences that may lead to noisy annotations, and the purple parts highlight the major differences between cases from the fact perspective or the judicial interpretation perspective.

extracting internal textual information to compare text similarity between the query and the candidate case, exemplified by methods like BERT-PLI [3] and SAILER [4]. These methods achieve better representations by leveraging the structural information of case texts. Although both approaches exhibit state-of-the-art performance, they all follow a data-driven paradigm, which means their efficacy heavily relies on clean and high-quality annotated data.

However, in practice, legal annotation requires highly specialized domain knowledge to identify nuances between cases. This complexity mainly arises from the inherent uncertainty in the legal domain, referred to as *legal uncertainty* [5–7], which represents the ambiguity or unpredictability in legal interpretations and applications. In the task of legal case retrieval, legal uncertainty can affect relevance estimation between two cases at the concept, rule, and fact levels, often leading to noisy correspondences and polluting the training data. In particular, the concept-level uncertainty indicates the ambiguity in the interpretation of legal concepts which often lack precise definitions and can be open to multiple interpretations. For example, as illustrated in Fig. 1, the term "justifiable defense" may be interpreted differently by different judges, lawyers, or annotators which induces incorrect labels. The rule-level uncertainty stems from the complex nature of legal statutes and regulations. For example, as shown in the middle of Fig. 1, although the two cases have different factual circumstances, they are relevant because they pertain to Article 309, which is open to multiple interpretations (Rule 1 and Rule 2). This variability can also result in incorrect annotations. The fact-level uncertainty arises from the challenges in establishing and interpreting the facts of a case. As illustrated at the bottom of Fig. 1, despite both cases involving a "stabbed," they are irrelevant because one suspect has the intent to kill, while the other does not. These three uncertainties significantly increase the difficulties of legal annotations, resulting in noisy correspondences in the relevance estimation of legal case retrieval. Unfortunately, existing studies on legal case retrieval [1–4,8] overlook the robustness against such noisy correspondences, as we will elaborate in the experiments section, thereby limiting their practical application.

Focusing on robustness against noisy correspondences, we propose a novel evidential learning framework called ELCR for the legal case retrieval task. This framework explicitly models the uncertainty caused by noisy correspondences and assigns correct labels, thereby improving retrieval performance under noisy conditions. Specifically, firstly, inspired by the legal uncertainty, we respectively estimate the query-candidate relevance from the concept, rule, and fact levels by incorporating knowledge from legal statutes. Secondly, we combine these relevance estimations to explicitly model relevance uncertainty by parameterizing relevance evidence into the Dirichlet distribution under Dempster-Shafer Evidence Theory [9]. The derived uncertainty helps correct observed labels potentially polluted by noisy correspondences. To address fluctuations in estimated uncertainty during the early stages of model training, we introduce a novel label correction strategy to prevent over-correction and stabilize learning. Thirdly, in addition to traditional relevance-oriented training objectives, we propose two evidence-based loss functions to enhance retrieval performance under noisy conditions. The one is an evidence-based contrastive learning loss, which ensures more evidence is generated for correct

predictions than for misclassifications. The other is an uncertainty penalty loss, which further regularizes the predictive distribution by penalizing divergences from uncertain retrievals. Extensive experiments on various noise proportions across two benchmark datasets demonstrate that our method exhibits robustness against noisy correspondences while maintaining competitive retrieval performance.

We summarize our contributions as follows:

- Inspired by the uncertainty in the legal domain, we propose an evidential learning framework, namely ELCR, to explicitly model multi-faceted relevance uncertainties and detect noisy labels. To the best of our knowledge, our work is the first effort aimed at improving robustness against noisy correspondence in legal case retrieval.
- We introduce an uncertainty-based strategy to correct noisy labels and stabilize the training process. Additionally, we devise two uncertainty-based learning objectives to ensure a desired evidence distribution, generating more evidence for correct classifications compared to misclassifications.
- Extensive experiments demonstrate that ELCR exhibits robustness against noisy correspondences while maintaining competitive retrieval performance.

The remainder of the paper is structured as follows. Sec. 2 offers a concise overview of previous research in legal case retrieval, uncertainty, and retrieval under noisy labels. Sec. 3 provides a comprehensive exposition of the proposed ELCR framework, detailing the multi-faceted legal relevance modeling, the uncertainty estimation, and the label correction strategy. Sec. 5 presents an in-depth analysis of ELCR through extensive experimental evaluation. Sec. 6 summarizes the key points of this paper.

2. Related work

In this section, we review the related work from two aspects: legal case retrieval, retrieval under noisy labels, legal uncertainty, and uncertainty-based learning.

2.1. Legal case retrieval

Legal case retrieval (LCR) is a long-standing research topic in the field of information retrieval. In contrast to general document retrieval, legal case retrieval involves a significant level of specialized legal knowledge and deals with lengthy and complicated legal documents. In the early exploration, researchers [10,11] made efforts to inject legal knowledge into the retrieval through the decomposition of legal issues. In recent years, deep learning demonstrated its effectiveness in exploring document semantics. One representative line of works focused on the network-based precedent methods tailored for the common law systems. For example, Minocha et al. [1] leveraged the Precedent Citation Network (PCNet) to predict the relevance based on whether the sets of precedent citations occur in the same cluster. Bhattacharya et al. [2] proposed Hier-SPCNet to capture all domain information inherent in laws and precedents. The other line of works judged the relevance between the query and candidate document according to their text-level similarity. One such method was BERT-PLI [3]. It divided the legal document into several paragraphs and used BERT [12] to obtain the similarity between the paragraphs. Lawformer [13] was another text-based method. It used millions of Chinese criminal and civil case documents to pre-train a Longformer [14] model. SAILER [4] was a novel method to handle structured case text. It introduced training objective that uses the fact description of the legal document to predict its judgment, expecting a proper alignment of legal documents in vector space based on their judgments. Despite impressive performance, these works rely heavily on clean and high-quality annotated data, while ignoring robustness against noisy correspondence.

2.2. Retrieval under noisy labels

In the context of noisy learning [15–17], previous researchers have proposed several robust training methods [18,15,16,19]. The vast majority of these methods are applied within Computer Vision (CV), with a smaller portion adapted for Natural Language Processing (NLP). Among these, some are tailored specifically for classification tasks and do not readily extend to the retrieval task. These methods can be broadly categorized into three classes: label correction, loss correction, and sample selection. Label correction approaches aim to enhance training robustness by rectifying erroneous labels [20,21]; however, they often necessitate additional clean data or a costly noise estimation process, which significantly curtails their practical utility. Loss adjustment methods bolster robustness through modifications to the loss function, such as BootStrap [22], ESC [16], and etc. Nonetheless, these designs are specifically devised for binary classification tasks, thereby inapplicable in listwise ranking scenarios. Regarding sample selection methodologies, many leverage the memorization characteristics of Deep Neural Networks (DNNs) to identify and utilize clean samples from amidst noisy training data for learning [23,24]. Yet, these techniques tend to be either exclusive to CV, specific to classification tasks, or require clean samples for semi-supervised training, failing to adequately address the challenges posed by noisy LCR settings. Recently, Marcuzzi et al. [25] proposed SOUR, a novel learning-to-rank method that enhances the model training process by detecting and eliminating outlier data points before constructing the ranking model. SOUR defined outliers in the ranking list through hyper-parameters, which required searching for the optimal configuration in a large parameter space in practical applications. We draw on its approach to selecting outliers in the ranking list and transform it into a comparison of different ranking distributions, reducing the number of hyper-parameters and enhancing the adaptability of the algorithm.

2.3. Legal uncertainty

Unlike fields dominated by determinism such as mechanical engineering, the legal field is characterized by inherent uncertainty [5,6]. Dari-Mattiacci and Deffains [26] pointed out the uncertainty in legal processes and elucidates the consequences caused by this uncertainty. Davis [7] distinguished between subjective and objective forms of legal uncertainty. The inherent uncertainty of this field, along with the diversity and complexity of real cases, makes it difficult for practitioners to make accurate judgments on certain cases based on precedent cases and the corresponding law. The previous works expounded the principles or applications of legal uncertainty across various domains of law. Nevertheless, they did not incorporate legal uncertainty into the realm of intelligent legal systems, nor did they leverage uncertainty estimations from machine learning to quantify legal uncertainty. Based on the previous theories of legal uncertainty, we consider the actual situation of legal case retrieval tasks and formalize the sources of legal noisy correspondences into three levels of uncertainty. Furthermore, we utilize evidential learning to estimate uncertainty, thereby identifying noise and selecting clean data for better training.

2.4. Uncertainty-based learning

In recent years, deep learning-based methods typically use a softmax layer for final output, but their deterministic predictions [27] and the overconfident tendency from the softmax layer [28] lead to an inability to assess the uncertainty of predictions like humans do. To measure uncertainty, some works adopt distributions instead of deterministic weight parameters [27,29], resulting in significant additional computational costs. Recently, Sensoy et al. [30] proposed Evidential Deep Learning (EDL), which utilizes the theory of Subjective Logic (SL) [31] to construct Dirichlet distributions similar to probabilities and learns functions based on evidence collected from data supporting these views. This method had gradually been applied in multi-modal tasks to obtain the uncertainty of cross-modal instances [32,33]. Wang et al. [34] focused on the text-to-text question classification task, drawing on the idea of EDL to capture the uncertainty of noisy data. Unlike these methods focusing on sample classification tasks, we measure uncertainty for text relevance ranking task to estimate the legal uncertainty inherent in the legal field itself.

2.5. Limitations of existing studies

Based on the literature review above, we summarize the limitations of existing studies in the field of legal case retrieval with noisy correspondences and highlight the novelty and advantages of our method:

- Despite the respective advantages, existing studies are not well-suited to the noisy scenarios in LCR for various reasons. On the one hand, state-of-the-art methods for LCR have not addressed the issue of noisy correspondence [35,8,36,13]. On the other hand, existing methods aimed at improving model robustness are either inapplicable due to task mismatch [15,19] or may result in performance deviations due to differing learning objectives [32,34].
- Moreover, legal uncertainty [5–7] is a key factor affecting the relevance of paired legal cases. Current research on modeling case relevance primarily relies on textual similarity, assuming that models will implicitly extract relevance signals. However, by overlooking the explicit modeling of legal uncertainty, this approach inevitably compromises the reliability of relevance predictions.

To tackle the above challenges and meet the demand for noise robustness in the context of LCR, our approach leverages the multi-faceted relevance of paired cases and employs evidential learning to quantify the uncertainty in their relevance estimation. This allows us to effectively identify and correct noisy samples, thereby improving the overall reliability and accuracy of legal case retrieval.

3. Methodology

In this section, we present our method designed for the task of legal case retrieval, improving the robustness against noisy correspondence.

3.1. Problem formulation

In this work, we focus on the legal case retrieval, which is formulated as follows. Given a set of samples $\mathcal{D} = \{(q, \mathcal{C}, \mathcal{R})\}$, each tuple in the dataset consists of a query case q, an undecided one submitted by the judge; $\mathcal{C} = \{c_1, c_2, \ldots, c_K\}$, a candidate precedent case set of size K; and $\mathcal{R} \subseteq \mathcal{C}$, a labeled relevant case set for query q. The objective of this task is to learn a retrieval function $f : q \times \mathcal{C} \rightarrow \mathcal{R}$ that takes (q, c) as input and maps the pair to a relevance score $r \in \mathbb{R}$. If $r > 0$ it indicates that $c \in \mathcal{R}$; otherwise $c \notin \mathcal{R}$. A common approach for the learning is to minimize a loss function that measures the disagreement between the relevance prediction $\hat{r}$ and the ground truth r^{gt}. Therefore, one key step for this task is estimating the relevance of (q, c) which we will elaborate on in Sec. 3.2.1.

As aforementioned, legal annotation requires specialized domain knowledge, which often introduces noise into the training data, resulting in noisy correspondences, i.e., r^{gt} may be corrupted by noise and we only observe the polluted labels r^{obs}. To simulate this annotation noise, we randomly select a certain proportion of data from the training set, denoted as $p \in \{10\%, 30\%, 50\%, 70\%, 90\%\}$. Then, we randomly swap the labels of positive and negative samples for the selected data. Therefore, in this study, we further aim to identify and correct these noisy labels r^{obs}, which will be elaborated in Sec. 3.2.2 and Sec. 3.2.3.

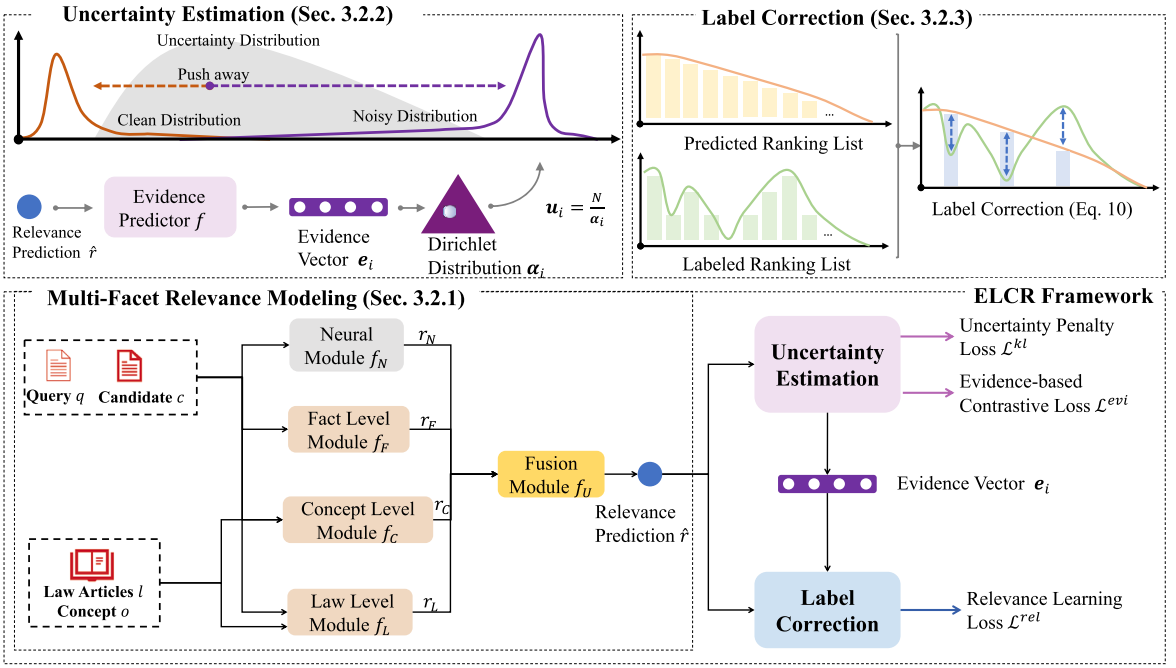


Fig. 2. The overall framework of ELCR includes three main components. The bottom-left illustrates the multi-faceted relevance modeling inspired by legal uncertainty. The top-left demonstrates explicit uncertainty modeling through evidential learning. The top-right showcases evidence-based noise detection.

3.2. The proposed framework

To address the issue of noisy correspondence in legal case retrieval, we enhance the existing neural module by introducing a novel evidential learning framework, referred to as ELCR. As illustrated in Fig. 2, ELCR first takes a query case q and a candidate case c as input and predicts their multi-faceted relevance $\hat{r}$. Subsequently, ELCR employs the Dempster-Shafer Theory of Evidence to model the query-candidate uncertainty u based on $\hat{r}$. Finally, ELCR detects noisy correspondence and corrects the observed label r^{obs} given the estimated uncertainty u.

3.2.1. Multi-faceted legal relevance modeling

The definition of legal relevance is quite complicated [35–37], requiring the identification of the nuances between query cases and candidate precedents. This process necessitates reference to various aspects of relevance, including the levels of legal concepts, law rules, and legal facts. Therefore, in this study, we introduce law knowledge to model the relevance between query and candidate cases from these three perspectives. This multi-faceted approach provides a comprehensive basis for subsequently modeling legal uncertainty and detecting noisy correspondences. Specifically, given a (q, c) pair, ELCR achieves this goal through the following five modules:

Neural Module f_N is responsible for predicting the relevance score from a semantic perspective for (q, c), which is formulated as:

$$r_N = f_N(q, c), \tag{1}$$

where $r_N \in \mathbb{R}$ denotes the degree of the relevance between q and c. Please note that f_N can be implemented using existing legal retrieval models, such as cross-encoder methods [13,3] or dual-encoder methods [38]. Therefore, ELCR is model-agnostic that can incorporate existing retrieval methods as f_N. In this study, we follow [39] and use Criminal-BERT [40] to implement f_N and we will discuss the implementation of f_N in the experiments.

Concept Level Module f_C aims to align the query case with the applicable laws corresponding to the candidate precedent from the perspective of legal concepts by incorporating law articles, where each legal concept is pre-collected following [39]. Specifically, f_C computes and aggregates the similarity between each sentence from the query case and the legal concepts as follows:

$$r_C = f_C(q, c) = \sum_i \max_j \left(\cos(s_i^q, o_j^c)\right), \tag{2}$$

where s_i^q denotes the i-th sentence in the query q; o_j^c denotes the j-th legal concept corresponding to the candidate c.

Law Level Module f_L works similarly to the concept-level module but directly aligns the query case with the applicable laws corresponding to the candidate precedent, i.e., f_L computes and aggregates the similarity between each sentence from the query case and the applicable laws:

$$r_L = f_L(q,c) = \sum_i \max_j \left(\cos(s_i^q, l_j^c)\right), \tag{3}$$

where s_i^q denotes the i-th sentence in the query q; l_j^c denotes the j-th applicable law corresponding to candidate c.

Fact Level Module f_F recognizes that certain parts of sentences in legal cases, such as criminal facts, contribute to the relevance measure, while others, such as the suspect profile, do not. Therefore, f_F aims to improve the neural module by computing the fine-grained similarity between the sentence from the query case and that from the candidate case as follows:

$$r_F = f_F(q,c) = \sum_i \max_j \left(\cos(s_i^q, s_j^c)\right), \tag{4}$$

where s_i^q and s_j^c respectively denotes the i-th and j-th sentence in the query q and candidate c.

Fusion Module f_U further combines the above relevance signals to ensure the quality of relevance modeling, given that f_F may produce spurious alignments, such as when the query and candidate have identical suspects. Concretely, f_U fuses the multi-faceted relevance signals and produce the relevance prediction $\hat{r}$ as follows:

$$\hat{r} = f_U\left(r_N, r_C, r_L, r_F\right), \tag{5}$$

where f_U is a linear projection layer. Please note that during training, we run the multi-faceted method above to effectively provide a basis for uncertainty estimation. Meanwhile, for efficiency purposes, we only utilize f_N during inference.

3.2.2. Uncertainty estimation

A critical challenge in learning from noisy correspondences is detecting such unreliable supervision signals. To address this, we employ the Dempster-Shafer Theory of Evidence (DST) to explicitly model the uncertainty in relevance estimation for legal case retrieval. DST offers a generalization of Bayesian probability theory and assigns belief masses to the set of mutually exclusive possible categories with one category expressing uncertainty itself. Following [30], we adhere to the principles of Subjective Logic [31] and formalize DST's belief assignments over a Dirichlet Distribution. For a K-way classification task, the belief masses b_k associated to K possible categories and the overall prediction uncertainty mass u are defined as follows:

$$u + \sum_1^K b_k = 1, \tag{6}$$

where $b_k \geqslant 0$ denotes the belief masses for k-th category; $u \geqslant 0$ denotes the uncertainty mass.

Intuitively, we conceptualize the retrieval as a classification problem [32,41,39,30], formulating the relevance prediction of a query-candidate pair as a classification task. Our evidential learning framework uses the uncertainty mass u to indicate ambiguity in the relevance prediction between a query and a candidate pair, thereby serving for detecting noisy correspondences in legal case retrieval. More precisely, given a batch of (q_i, c_j) and their multi-faceted relevance prediction $\hat{r}_{ij}$ with batch size B, the corresponding relevance belief and uncertainty are:

$$b_{ij} = \frac{e_{ij}}{L_i} = \frac{\alpha_{ij} - 1}{L_i}, \quad u_i = \frac{B}{L_i}, \tag{7}$$

where $e_{ij} = f_E(\hat{r}_{ij}) = \exp^{\tanh\left(\hat{r}_{ij}/\tau\right)}$ is the relevance evidence denoting the measure of the amount of support for being classified into a certain relevance level based on Sec. 3.2.1, f_E denotes the evidence predictor; $0 < \tau < 1$ denotes the temperature parameter; $L_i = \sum_{j=1}^B (e_{ij} + 1)$ denotes the strength of Dirichlet distribution; the belief $\boldsymbol{b_i} = \left[b_{ij}\right]_{j=1}^B$ can be regarded as the subjective opinion obtained from the parameters of the corresponding Dirichlet distribution $\boldsymbol{\alpha_i} = \left[\alpha_{ij}\right]_{j=1}^B$. Unlike traditional approaches that involve neural networks to produce probability assignments for each possible class, we leverage the evidence-parameterized Dirichlet distribution to represent the density of these class assignments, thereby modeling second-order probabilities and uncertainty [31]. Consequently, our method possesses the advantage of detecting noisy correspondences effectively. The density function characterized by $\boldsymbol{\alpha_i}$ is defined as:

$$D\left(\hat{\boldsymbol{r}}_i \mid \boldsymbol{\alpha}_i\right) = \begin{cases} \frac{1}{Beta(\boldsymbol{\alpha}_i)} \prod_{j=1}^B \hat{r}_{ij}^{\alpha_{ij}-1} & \text{for } \hat{\boldsymbol{r}}_i \in S_B \\ 0 & \text{otherwise} \end{cases}, \tag{8}$$

where $\hat{\boldsymbol{r}}_i \in \mathbb{R}^B$ are the query probabilities, $Beta\left(\boldsymbol{\alpha}_i\right)$ is the B-dimensional multinomial beta function, and S_B is the B-dimensional unit simplex [42].

3.2.3. Label correction

Due to the inherent complexity of legal annotation, the observed relevance labels in legal case retrieval typically involve noisy correspondences, leading to unreliable supervision signals. Mainstream solutions for this task follow a pre-train then fine-tune paradigm, leveraging knowledge obtained during the pre-training phase. However, in practical scenarios, there often exists a distribution shift between the clean pre-training corpus and the noisy fine-tuning data. This discrepancy increases the uncertainty in relevance predictions, compromising the model's confidence in its outputs and inevitably resulting in degraded accuracy in legal case retrieval. To address these challenges, we utilize the derived relevance evidence and uncertainty discussed in the preceding sections to correct the

observed labels. Specifically, we focus on correcting labels with high uncertainty, which is inversely proportional to the evidence, as follows:

$$r_{ij}^{c} = \begin{cases} r_{ij}^{o} & \text{if } j = \arg\max\left(\boldsymbol{e}_i\right) \\ 0 & \text{else} \end{cases}, \tag{9}$$

where $\boldsymbol{e}_i \in \mathbb{R}^B$ is the relevance evidence vector, r_{ij}^{o} is the observed label of (q_i, c_j). Given the multi-faceted relevance evidence estimations from the concept, rule, and fact levels, our correction method effectively identifies pairs with noisy correspondences. Additionally, considering that pre-trained models can achieve remarkable performance with only a few clean training samples [35,8], we treat these noisy pairs as negative examples, i.e., assign the $r_{ij}^{c} = 0$ for convenience.

Despite its efficacy, the correction strategy in Eq. (9) is aggressive, as it relies on the current relevance evidence. A common scenario is that, in the early stages of training, the predictions of relevance evidence may fluctuate significantly. This, in turn, leads to fluctuating corrected labels, resulting in an unstable training process. Therefore, we implement a warm-up phase wherein the proposed framework initially learns from noisy labels for w_1 epochs. Following this, we correct these supervision labels based on the relevance evidence. In other words, from epoch $w_1 + 1$ onwards, our framework transitions to learning from the corrected labels. Similarly, to prevent the over-correction issue and further stabilize the training, we improve Eq. (9) increasing the trust in the model's predictions after epoch w_2 as follows:

$$\bar{r}_{ij} = \begin{cases} r_{ij}^{c} & \text{if } |r_{ij}^{c} - r_{ij}^{o}| \geq \frac{1}{B}\sum_j |r_{ij}^{c} - r_{ij}^{o}| \\ \hat{r}_{ij} & \text{else} \end{cases}. \tag{10}$$

3.3. Training

To achieve robust legal case retrieval with noisy correspondence, our ELCR has three training objectives to determine its parameters as follows:

Relevance learning. As the primary goal of legal case retrieval is to predict the relevance between query-candidate pairs, our first learning objective is to minimize the discrepancy between the corrected relevance labels and those predicted by our ELCR. Following [39], we apply the MSE loss function to achieve this goal:

$$\mathcal{L}_i^{rel} = \frac{1}{B}\sum_{j=1}^{B}\left(\hat{r}_{ij} - \bar{r}_{ij}\right)^2. \tag{11}$$

Evidence-based contrastive learning. The essence of contrastive learning [32,39] lies in learning effective representations of data by comparing and contrasting the positive and negative data points. Inspired by this, we implement contrastive learning from the evidence perspective to generate more evidence for the correct relevance prediction than that for misclassification. Formally, given a batch of query-candidate pairs with size B, we view the retrieval as B-way classification task. For each pair (q_i, c_j), the corresponding corrected the relevance labels $\bar{\boldsymbol{r}}_i = \left[\bar{r}_{ij}\right]_{j=1}^{B}$ are formulated as a B-dimensional vector. Since we explicitly model the relevance evidence e_{ij}, we can directly minimize the discrepancy between the query probabilities $\hat{\boldsymbol{r}}_i = \left[\hat{r}_{ij}\right]_{j=1}^{B}$ and the corrected labels $\bar{\boldsymbol{r}}_i$ by optimizing the squared error $\left\|\hat{\boldsymbol{r}}_i - \bar{\boldsymbol{r}}_i\right\|_2^2$ as follows:

$$\begin{aligned}
\mathcal{L}_i^{evi} &= \int \left\|\hat{\boldsymbol{r}}_i - \bar{\boldsymbol{r}}_i\right\|_2^2 \frac{1}{Beta\left(\boldsymbol{\alpha}_i\right)} \prod_{i=1}^{B} \hat{r}_{ij}^{\alpha_{ij}-1} d\hat{\boldsymbol{r}}_i \\
&= \sum_{j=1}^{B} \mathbb{E}\left(\bar{r}_{ij}^2 - 2\bar{r}_{ij}\hat{r}_{ij} + \hat{r}_{ij}^2\right) \\
&= \sum_{j=1}^{B} \left(\bar{r}_{ij}^2 - 2\bar{r}_{ij}\mathbb{E}\left(\hat{r}_{ij}\right) + \mathbb{E}\left(\hat{r}_{ij}^2\right)\right) \\
&= \sum_{j=1}^{B} \left[\left(\bar{r}_{ij} - \mathbb{E}\left(\hat{r}_{ij}\right)\right)^2 + \mathrm{Var}\left(\hat{r}_{ij}\right)\right] \\
&= \sum_{j=1}^{B} \left[\left(\bar{r}_{ij} - \frac{\alpha_{ij}}{L_i}\right)^2 + \frac{\alpha_{ij}\left(L_i - \alpha_{ij}\right)}{L_i^2\left(L_i + 1\right)}\right],
\end{aligned} \tag{12}$$

where the density of $\hat{\boldsymbol{r}}_i$ follows the parameterized Dirichlet distribution $\boldsymbol{\alpha}_i$ [32,41], $\mathbb{E}(\hat{r}_{ij})$ and $\mathrm{Var}(\hat{r}_{ij})$ represent the expected value and the variance of $\hat{r}_{ij}$, respectively. As shown in Eq. (12), for a given query q_i with the relevant candidate c_j, this loss helps approach $\mathbb{E}(\hat{r}_{ij})$ closer to $\bar{r}_{ij}$ while reducing the variance. This empowers the optimized neural network to generate more evidence for correct classification and helps avoid misclassification by reducing misleading evidence and decreasing the variability in its predictions [30].

Uncertainty penalty. The evidence-based contrastive learning helps in the evidence assignment, making the evidence for the correct prediction higher than that for misclassification. However, this objective cannot ensure the total evidence shrinks to zero if one

sample cannot be correctly classified. To fix this issue, we further introduce a Kullback-Leibler (KL) divergence term as a regularization term. It regularizes our predictive distribution by penalizing divergences from the uncertain retrieval, which is formulated as:

$$\begin{aligned}\mathcal{L}_i^{kl} &= \mathrm{KL}\left[D\left(\hat{\boldsymbol{r}}_i \mid \tilde{\boldsymbol{\alpha}}_i\right) \| D\left(\hat{\boldsymbol{r}}_i \mid \mathbf{1}\right)\right] \\ &= \log\left(\frac{\Gamma\left(\sum_{j=1}^{B} \tilde{\alpha}_{ij}\right)}{\Gamma(B)\prod_{j=1}^{B}\Gamma\left(\tilde{\alpha}_{ij}\right)}\right) + \sum_{j=1}^{B}\left(\tilde{\alpha}_{ij}-1\right)\left[\psi\left(\tilde{\alpha}_{ij}\right) - \psi\left(\sum_{j=1}^{B}\tilde{\alpha}_{ij}\right)\right],\end{aligned} \tag{13}$$

where $\tilde{\boldsymbol{\alpha}}_i = \bar{\boldsymbol{r}}_i + \left(\mathbf{1} - \bar{\boldsymbol{r}}_i\right) \odot \boldsymbol{\alpha}_i$ is the Dirichlet parameters after removing the unreliable evidence from the predicted Dirichlet distribution; $D\left(\hat{\boldsymbol{r}}_i \mid \mathbf{1}\right)$ is the uniform Dirichlet distribution, and $\Gamma()$ and $\psi()$ are the gamma and digamma functions, respectively.

Therefore, the overall training objective is summarized as follows:

$$\mathcal{L} = \sum_{i \in D} \mathcal{L}_i^{rel} + \lambda_{evi}\mathcal{L}_i^{evi} + \lambda_{kl}\mathcal{L}_i^{kl}, \tag{14}$$

where λ_{evi} and λ_{kl} are the balancing coefficients.

3.4. Inference

Following the practice of [32], we assume the given cleaning test sets are clean. Therefore, distinct from the training phase, during inference, only the Neural Module of our ELCR in Eq. 2 is activated, without engaging in uncertainty estimation or label correction. This is also rooted in the consideration of inference efficiency. It is noteworthy that we assume the knowledge mined by the other modules has already been integrated into the Neural Module through training.

3.5. Computational complexity analysis

The time overhead of ECLR comes from the following components: Evidence Calculation and Uncertainty Estimation, Label Correction and Relevance Prediction, and the Backbone Network Forward Pass. We analyze the time complexity of each module as follows:

- **Evidence Calculation and Uncertainty Estimation:** The complexity of calculating the evidence vector $\mathbf{e}_i$ and belief $\mathbf{b}_i$ from the Dirichlet distribution is $O(B)$ per query-candidate pair. Given N pairs, the overall complexity is $O(N \cdot B)$, where B is the size of mini-batch.
- **Label Correction and Relevance Prediction:** For each query-candidate pair, the label correction step involves selecting the maximum evidence among B relevance classes, yielding a complexity of $O(B)$ per pair, i.e., $O(N \cdot B)$ for the whole dataset.
- **Backbone Network Forward Pass:** The backbone network introduces a complexity of $O(D^2)$ per query-candidate pair, where D is the hidden dimension of the neural network. For a batch of N pairs, this results in $O(N \cdot D^2)$.

The derived overall time complexity of our ECLR is $O(N \cdot (B + D^2))$, and in most case, we have $B \ll D^2$. Thus, the overall computational complexity of our ECLR is $O(N \cdot D^2)$, where N is the number of query-candidate pairs and D is the hidden dimension of the backbone neural network. Considering that all popular LCR methods adopt transformer architectures with a time complexity of $O(N \cdot D^2)$, our ECLR is comparable to these methods in terms of computational complexity.

4. Discussion

In this part, we highlight the merits and limitations of the proposed ELCR approach as follows:

Merits 1: Robust Retrieval Performance Against Noisy Correspondences. Existing LCR methods heavily rely on clean annotated data for effective retrieval performance and do not account for the practical issue of noisy correspondences that often arise in real-world LCR annotation. In contrast, to the best of our knowledge, ELCR is the first method specifically designed to address this challenge. By employing evidential learning for noise identification and leveraging a multi-faceted understanding of similarity in legal cases, ELCR significantly improves retrieval performance in the presence of noise.

Merits 2: Explicit Modeling of Legal Uncertainty. Current LCR methods, such as BERT-ts-L1 [36] and Lawformer [13], overlook the modeling of legal uncertainty, limiting their ability to handle ambiguous or confusing relevance judgments. This overconfidence in the annotations can lead to poor performance, particularly when dealing with noisy or ambiguous correspondences. ELCR introduces an evidential learning framework that explicitly models legal uncertainty in relevance estimation. By parameterizing relevance evidence into a Dirichlet distribution, ELCR can quantify the level of uncertainty for each query-candidate pair, enabling more effective label correction and reducing overfitting to noisy data. This uncertainty-aware approach is a key factor in ELCR's robustness.

Merits 3: Adaptability and Flexibility. Another strength of ELCR is its adaptability to different backbone models. In the following experiments, we demonstrate that ELCR can be easily integrated with various neural architectures, such as BERT [12], CriminalBERT [40], and Lawformer [13]. This flexibility allows ELCR to leverage the strengths of existing retrieval models while adding robustness to noisy correspondences and uncertainty estimation. In contrast, most current LCR methods are tightly coupled to a

Table 1
Statistics of LeCaRD and ELAM.

	LeCaRD	ELAM
# total queries	107	85
# candidate cases per query	100	50
Avg. # relevant cases per query	10.33	9.06
Avg. # sentences per query	6.91	16.24
Avg. # sentences per candidate case	86.63	43.43
Avg. # cited law articles per candidate case	6.5	5.62

specific architecture and may not generalize well across different models or retrieval tasks. ELCR's modular design makes it a versatile framework that can be applied to a wide range of legal case retrieval tasks with minimal modifications.

Limitation 1: Potential Reduction in Positive Sample Availability. In scenarios with a large amount of noise, ELCR's inability to perfectly differentiate between noisy positive samples and true negatives may result in a reduction in the number of available positive samples. This can potentially lead to a decrease in model performance, especially under conditions of high noise levels. One potential solution to this issue is to design additional modules specifically for high-noise situations to identify potential positive sample pairs from those labeled as 0.

Limitation 2: Computational Cost. While the evidential learning framework is effective for managing uncertainty and noise, it inevitably introduces additional computational overhead. The extra steps involved in evidence calculation, uncertainty estimation, and label correction increase computational demands, which may pose challenges in resource-constrained environments.

5. Experiments

In this section, we conduct extensive experiments to answer the following research questions:

- **RQ1:** How does ELCR perform on legal case retrieval under varying noise strengths compared to state-of-the-art methods?
- **RQ2:** How effective is each module within ELCR?
- **RQ3:** How well can ELCR detect noisy correspondence given the polluted observed relevance labels?
- **RQ4:** Can our ELCR be adapted to other backbone models, i.e. f_N?

5.1. Experimental settings

5.1.1. Datasets and evaluation metrics

To comprehensively evaluate the performance of our method, we conduct experiments on two benchmark datasets of legal case retrieval.

- **LeCaRD** [35] is a legal case retrieval dataset that contains 107 queries and 43,000 target cases. For each query, there are 100 target cases provided as candidates. Each target case is assigned a 4-level relevance label, which indicates the degree of relevance between the query and the case. In the candidate pool, about 70% of the candidates are negatives. All the criminal cases included in this dataset were published by the Supreme People's Court of China, which is the highest judicial body in China.
- **ELAM** [38] is an explainable legal case matching dataset, covering about 5K pairs of annotated legal cases. Each pair of cases was manually assigned a label, which was categorized into three levels, ranging from 3 to 0, indicating a gradual decrease in relevance. Following the setup of Sun et al. [39], we process ELAM into legal case retrieval format. Each query will have a candidate pool composed of 50 candidate examples, including both positive and negative samples.

The statistics of LeCaRD and ELAM are shown in Table 1. ELAM and LeCaRD respectively have 3 and 4 levels of relevance labels where $\{0, 1, 2\}$ denotes irrelevant, partially relevant, and relevant in ELAM; $\{1, 2, 3, 4\}$ indicates a gradual decrease in the degree of relevance in LeCaRD. Following the setup of [39], we normalize the relevance labels in LeCaRD and ELAM, which have multiple-level relevance labels to ensure that the ground-truth relevance scores range from 0 to 1. To reduce the impact of randomness, we conduct five repetitions for each experiment and report the average values.

5.1.2. Baselines

Learning-to-rank baselines In this study, we evaluate several advanced models as baselines for the task of legal case retrieval. These models include Criminal-BERT, BERT-ts-L1, Lawformer, SAILER, SOUR, Condenser, and coCondenser.

- **Condenser** [43] is designed for ad-hoc dense retrieval, using skip connections to aggregate textual information into dense vectors. This mechanism enables Condenser to capture and represent the essential information of legal texts efficiently.
- **coCondenser** [44] extends Condenser by incorporating an unsupervised corpus-level contrastive loss. This addition helps warm up the vector space effectively, improving the model's performance in dense retrieval tasks by enhancing the quality and consistency of the generated vectors.

- **Criminal-BERT** [40] is a specialized pre-trained model based on BERT and fine-tuned on millions of criminal legal cases. This model utilizes a cross-encoder architecture to predict the relevance between a query and candidate cases, leveraging extensive legal data to improve its understanding of legal terminology and case specifics.
- **BERT-ts-L1** [36] builds upon Criminal-BERT by optimizing its attention weights with feedback from legal experts. This user-guided optimization refines the model's ability to predict the relevance of legal cases, enhancing its retrieval accuracy in legal contexts.
- **Lawformer** [13] is a pre-trained language model based on Longformer [14], designed to process lengthy legal cases effectively. Trained on millions of Chinese legal cases, Lawformer represents long legal texts efficiently. In our experiments, we concatenate input cases and use mean pooling of Lawformer's output for document matching, ensuring comprehensive utilization of input data.
- **SAILER**[1] [4] is a structure-aware pre-trained language model that utilizes the structural information inherent in legal cases and employs an asymmetric encoder-decoder architecture to integrate multiple pre-training objectives, encoding rich semantic information into dense vectors.
- **SOUR**[2] [25] contributes to the learning-to-rank scenario by addressing the issue of outliers, thereby improving the ranking of legal cases with noisy associations. Outliers, which can negatively impact ranking models, are effectively mitigated by an innovative approach introduced by SOUR. This approach involves the detection and removal of these outliers prior to the construction of the ranking model, which substantially enhances the model's reliability and precision in legal case ranking.

Uncertainty-based denoising baselines To conduct a more comprehensive comparison, we introduce additional robust methods beyond the learning-to-rank scenario. Due to differences in setup, most of these methods cannot be directly applied to the legal case retrieval task. Therefore, we have made certain adjustments to the settings of each method.

- **DBAESS** [18] is a divergence-based approach with enhanced sample selection to address the common challenges of label noise and distribution imbalance in real-world image classification scenarios and mitigate the adverse effects of these issues. Since our data is textual, we adopt Criminal-BERT as the text encoder, maintaining consistency with other baselines. We use a fully connected layer to obtain the predicted probability distribution (classifying into 3 and 4 categories for the ELAM and LeCaRD datasets, respectively). We distinguish noisy samples according to the method provided in the paper. In the second stage, we follow the paper's approach, adapting semi-supervised learning to the text scenario.
- **ERDL** [34] is an Evidential Robust Deep Learning (ERDL) framework that integrates contrastive and evidential learning losses to accurately capture semantic similarity and handle noisy data. Since this work targets text classification problems, we can directly simplify our task into a classification task, maintaining consistency with the processing we did in DBAESSS. Meanwhile, we use Criminal-BERT as the text encoder.
- **IEP** [18] is a framework for combining multiple Conformal Predictors (CPs) inspired by possibilistic information fusion. By treating the p-value functions of CPs as possibility distributions and applying combination rules, this method enhances classification accuracy and robustness. To adapt this method to our scenario, we obtain the possibility distributions for sample pairs through two Criminal-BERT models and then follow the combination rules (weighted IEP) that performed the best in their paper.
- **ESC** [16] refers to Equivariant Similarity Consistency. It enhances model robustness against noisy data by comparing semantic variations between images and texts in response to the noisy correspondence issue in cross-modal data matching. To adapt this method to our scenario, we change the original image-text pairs setting to text pairs setting and replace the image encoder with a text encoder, specifically Criminal-BERT. Consistent with ELCR, we still frame it as a regression task to meet the requirements of the similarity ranking task.

5.1.3. Evaluation metrics

Following [35], we adopt precision metrics like Precision, Mean Average Precision (MAP), and Normalized Discounted Cumulative Gain (NDCG) to assess the performance of baselines and our ECLR. Formally, NDCG is defined as follows:

$$\text{NDCG@}k = \frac{\text{DCG@}k}{\text{IDCG@}k},\ \text{DCG@}k = \sum_{i=1}^{k} \frac{r_i}{\log_2(i+1)}, \tag{15}$$

where k denotes the position in the ranked list, and IDCG@k denotes represents the ideal DCG score at position k. In our experiments, we evaluate NDCG for $k \in \{10, 20, 30\}$. To demonstrate a more intuitive comparisons, we also introduce "Avg NDCG" and "Avg Metrics", representing the average value of NDCG and the average of all six metrics, respectively.

5.1.4. Implementation details

Since our ELCR is a model-agnostic framework that can easily incorporate existing neural models into f_N for legal case retrieval, we follow the practice [39] and select Criminal-BERT as the backbone of ELCR. We will also discuss the robustness of the backbone model in the experiments. For baselines and ours, we adopt the open-sourced checkpoints and codes to warm up model parameters and fine-tune them on ELAM and LeCaRD. We optimize the hyper-parameters through grid search on the validation set, employing

[1] https://github.com/CSHaitao/SAILER.
[2] https://github.com/FedericoMarcuzzi/Filtering-out-Outliers-in-Learning-to-Rank.

Table 2
Performance decline with increasing noise proportion magnitude. The best performances are denoted in bold fonts. "Avg. NDCG" and "Avg. Metrics" refers to the average value of NDCG and the all six metrics, respectively.

Dataset		LeCaRD			ELAM		
Increasing Noise Proportion Magnitude		10%→50%	10%→70%	10%→90%	10%→50%	10%→70%	10%→90%
Avg. Metrics	All Methods	-17.82%	-42.21%	-49.22%	-18.98%	-46.86%	-53.33%
	Baselines	-21.62%	-45.27%	-53.24%	-23.12%	-49.83%	-59.67%
	ELCR	**-5.81%**	**-32.52%**	**-36.49%**	**-5.17%**	**-36.97%**	**-32.20%**
MAP	All Methods	-4.03%	-8.22%	-6.63%	-17.87%	-37.81%	-45.89%
	Baselines	**-3.90%**	-8.59%	-7.21%	-20.29%	-40.56%	-51.21%
	ELCR	-4.46%	**-7.01%**	**-4.68%**	**-9.56%**	**-28.37%**	**-27.68%**
Avg. NDCG	All Methods	-26.17%	-62.79%	-75.01%	-19.85%	-53.95%	-59.16%
	Baselines	-32.55%	-67.91%	-81.65%	-25.35%	-57.16%	-66.36%
	ELCR	**-6.58%**	**-47.06%**	**-54.63%**	**-1.85%**	**-43.46%**	**-35.61%**

Adam. The batch sizes for all other models are set to 30. The max length of the model's input is set 4096 for Lawformer and 512 for other models. The batch size is set to 30, aside from Lawformer having its batch size set to 2 under the memory constraint. The learning rate is chosen from {1e-5, 2e-6, 4e-7}. λ_{evi} and λ_{kl} are tuned from [0.1, 0.5] and [0.05, 0.1], respectively. w_1 and w_2 are tuned from [1, 20]. τ is set to 0.1. ALL experiments are conducted on a single NVIDIA RTX A6000.

5.2. Retrieval performance under noisy correspondences

We first conduct extensive experiments to simulate various proportions of noisy correspondences in legal case retrieval. We first set p to 10%, 30%, 50%, 70%, and 90% respectively, to test the retrieval performance of the baselines and ours under different levels of noise intensity. The performances are shown in Table 2, Table 4, and Table 3. Table 2 summarizes the detailed experimental results from Table 4 and Table 3, demonstrating the changes in the experimental results with the noise intensity increasing. From the results, we have the following observations for **RQ1**:

5.2.1. The effects of noise intensity

As shown in Table 2, we find that as the noise intensity gradually increases for both the LeCaRD and ELAM datasets, the retrieval performance degradation of the various methods exhibits an increasingly aggravating trend, with specific decline magnitudes. Specifically, taking the "10%→90%" scenario as an example, for the LeCaRD dataset, the methods demonstrate noise robustness in MAP, with their performance in these metrics only declining by 6.63% overall as the noise proportion increases from 10% to 90%. However, the methods suffer severe harm in the NDCG metric, with their NDCG performance declining by 75.01% overall under the same noise increase. For the ELAM dataset, the methods experience significant degradation across all metrics, with an overall 45.89% decline in P@5, P@10, and MAP, and a 62.05% decline in NDCG metrics.

These results validate the detrimental impact of noisy correspondence in the legal case retrieval scenario, with the harm being most pronounced in the "Avg. NDCG" that is most relevant to the multi-label retrieval task.

5.2.2. Noise robustness comparison between baselines and ELCR

As shown in Table 2, in the vast majority of conditions, ELCR can significantly reduce the damage caused by noise compared to other baselines. ELCR can reduce the overall decline in each indicator under the three noise proportion change conditions in Table 2 by an average of approximately 41.68%. It is worth noting that under the noise proportion change condition of "10% → 50%", ELCR reduces the damage of noise to the "Avg. NDCG" on the LeCaRD and ELAM datasets by 79.78% and 92.70%, respectively. ELCR shows competitive performance under low-noise conditions and exhibits a significant advantage under high-noise conditions. This advantage increases with the increase in noise proportion. Specifically, except for the condition with 10% added noise on ELAM, ELCR outperforms all other methods in terms of both "Avg. NDCG" and "Avg. Metrics". Even though in some conditions (such as 10% noise proportion) and for some metrics (such as MAP), where ELCR cannot outperform certain baselines (such as Lawformer and SOUR), ELCR still surpasses other models in overall performance.

Detailed experimental results are shown in Table 4 and Table 3. These results further verify the effectiveness of our proposed ELCR.

5.2.3. Noise robustness from the perspective of pre-training-fine-tuning distribution gap

The baselines can be divided into two categories: one category is the methods that have not been pre-trained with legal corpora, such as SOUR, Condenser, and coCondenser; the other category is the methods that use legal corpora for pre-training, such as SAILER, Lawformer and so on. We find that the methods with legal pre-training have poorer noise robustness compared to the methods without legal pre-training.

Specifically, as shown in Table 4 and Table 3, under low noise conditions, the legal pre-training methods like Lawformer exhibit very strong retrieval performance. For example, it outperforms other models on the 10% noise ELAM dataset. However, this category of methods suffers greater damage from increased noise compared to the methods without legal pre-training. For example, on the ELAM

Table 3
Performance comparisons of our approach and the baselines on LeCaRD. The best and the second-best performances are denoted in bold and underlined fonts, respectively. "Ratio" refers to the proportion of noise within the candidate pool. "Avg. NDCG" and "Avg. Metrics" refers to the average value of NDCG and the all six metrics, respectively. Paired t-tests are conducted and "†" indicates the best performances are statistically significant compared with the second-best (p-value < 0.05).

Ratio	Method	P@5	P@10	MAP	NDCG@10	NDCG@20	NDCG@30	Avg. NDCG	Avg. Metrics
10%	condenser	0.3400	0.3800	0.4068	0.5538	0.5338	0.5629	0.5502	0.4629
	co-codenser	0.3900	0.3950	0.4351	0.4650	0.4548	0.4773	0.4657	0.4362
	Lawformer	**0.4580**†	**0.4250**†	0.4862	0.6468	0.6665	0.7044	0.6726	0.5648
	SAILER	0.3700	0.3550	0.4286	0.6657	0.7009	0.7258	0.6975	0.5410
	BERT-ts-L1	0.3600	0.3750	0.4753	0.6929	0.7390	0.7743	0.7354	0.5694
	Criminal-BERT	0.4100	0.3700	0.4795	0.7034	0.7407	**0.7817**†	0.7419	0.5792
	SOUR	0.3600	0.3450	0.4152	0.6655	0.6968	0.7460	0.7028	0.5381
	ELCR	0.3900	0.3900	**0.5075**†	**0.7107**†	**0.7453**†	0.7766	**0.7442**†	**0.5867**†
30%	condenser	0.3700	0.3700	0.4163	0.4373	0.4645	0.4965	0.4661	0.4258
	co-codenser	0.3100	0.3400	0.4016	0.4298	0.4364	0.4788	0.4483	0.3994
	Lawformer	0.3800	0.3650	0.4303	0.5022	0.5210	0.5697	0.5310	0.4614
	SAILER	0.3700	**0.3850**†	0.4309	0.6012	0.6162	0.6399	0.6191	0.5072
	BERT-ts-L1	0.3400	0.3550	0.4232	0.6692	0.7142	0.7510	0.7115	0.5421
	Criminal-BERT	**0.3900**†	0.3500	0.4347	0.6817	0.7263	0.7566	0.7215	0.5566
	SOUR	0.3700	0.3500	0.4099	0.6475	0.6671	0.7137	0.6761	0.5280
	ELCR	0.3700	0.3550	**0.4841**†	**0.6828**†	**0.7299**†	**0.7608**†	**0.7245**†	**0.5638**†
50%	condenser	0.3100	0.3500	0.4049	0.3118	0.3463	0.3943	0.3508	0.3529
	co-codenser	0.3200	0.3650	0.4209	0.3102	0.3508	0.3980	0.3530	0.3608
	Lawformer	0.3500	0.3850	0.4275	0.3256	0.3421	0.3802	0.3493	0.3684
	SAILER	**0.4300**†	**0.3900**†	**0.4756**†	0.3767	0.3983	0.4406	0.4052	0.4185
	BERT-ts-L1	0.3800	0.3550	0.4281	0.5395	0.5559	0.5765	0.5573	0.4725
	Criminal-BERT	0.3600	0.3850	0.4114	0.5562	0.5839	0.5993	0.5798	0.4827
	SOUR	0.3800	0.3750	0.4189	0.5126	0.4876	0.5032	0.5011	0.4462
	ELCR	0.3700	0.3750	0.4456	**0.6662**†	**0.7019**†	**0.7354**†	**0.7012**†	0.5490
70%	condenser	0.3400	0.3250	0.4006	0.2366	0.2647	0.3028	0.2680	0.3116
	co-codenser	0.3300	0.3300	0.4269	0.3372	0.3364	0.3723	0.3486	0.3555
	Lawformer	0.3000	0.3150	0.4029	0.2194	0.2339	0.2585	0.2373	0.2883
	SAILER	0.3000	0.3350	0.3810	0.1182	0.1409	0.1750	0.1447	0.2417
	BERT-ts-L1	0.3600	0.3450	0.4006	0.0798	0.1209	0.1376	0.1128	0.2407
	Criminal-BERT	0.3700	0.3450	0.4322	0.1850	0.1832	0.1968	0.1883	0.2854
	SOUR	**0.4300**†	**0.3900**†	**0.4667**†	0.1687	0.1618	0.1661	0.1655	0.2972
	ELCR	0.4200	**0.3900**†	0.4102	**0.3722**†	**0.3958**†	**0.4433**†	**0.4038**†	**0.4053**†
90%	condenser	0.3300	0.3500	0.4262	0.1367	0.1521	0.1696	0.1528	0.2608
	co-codenser	0.3300	0.3400	0.4125	0.1486	0.1547	0.1784	0.1606	0.2607
	Lawformer	0.3500	0.3300	**0.4353**	0.1046	0.1381	0.1643	0.1357	0.2537
	SAILER	0.3300	0.3350	0.3811	0.0993	0.0968	0.1101	0.1021	0.2254
	BERT-ts-L1	0.3900	0.3600	0.4099	0.0748	0.0902	0.1006	0.0885	0.2376
	Criminal-BERT	0.3700	0.3600	0.4313	0.0639	0.0630	0.0786	0.0685	0.2278
	SOUR	0.3700	**0.3800**	0.4207	0.1218	0.1383	0.1297	0.1299	0.2601
	ELCR	**0.4100**†	0.3600	0.4252	**0.3102**†	**0.3427**†	**0.3912**†	**0.3480**†	**0.3732**†

dataset, as the noise increases from 10% to 50%, Lawformer's average NDCG declines by 0.3415, while Condenser and coCondenser only decline by 0.1140 and 0.0362 respectively. This results in Lawformer performing more than 0.1 lower in average NDCG compared to coCondenser under 50% noise proportion on ELAM, whereas originally Lawformer was the top-performing method under 10% noise proportion condition on ELAM, while coCondenser was the poorest performer.

On one hand, we assume that the reason for this phenomenon is that these methods have a greater capability to capture more information from the legal text (for example, Lawformer can acquire longer text content, and SAILER integrates legal case structure knowledge). The specialized designs of these methods enhance the models' fitting capabilities, but under noisy conditions, they are more prone to fitting noise correlations, making them more susceptible to noise influence.

On the other hand, ELCR demonstrates strong noise robustness, even when using models pre-trained on legal corpora. This robustness can be attributed to the evidence-based label correction mechanism, which is sensitive to distribution shifts and effectively detects and corrects noisy labels.

5.3. Ablation study

To answer **RQ2** and better illustrate the effectiveness of each component of ELCR, we further conduct ablation studies on LeCaRD and ELAM. Table 5 demonstrates, using 50% noise as an example, the performance of our method without each component. We find that ELCR outperforms all variant models in the ablation study, and each variant also surpasses the best baseline, demonstrating the

Table 4
Performance comparisons of our approach and the baselines on ELAM. The best and the second-best performances are denoted in bold and underlined fonts, respectively. "Ratio" refers to the noisy correspondence proportion of within the candidate pool. "Avg. NDCG" and "Avg. Metrics" refers to the average value of NDCG and the all six metrics, respectively. Paired t-tests are conducted and "†" indicates the best performances are statistically significant compared with the second-best (p-value < 0.05).

Ratio	Method	P@5	P@10	MAP	NDCG@10	NDCG@20	NDCG@30	Avg. NDCG	Avg. Metrics
10%	condenser	0.4706	0.5059	0.5086	0.4611	0.5838	0.6803	0.5751	0.5351
	co-codenser	0.4706	0.4353	0.4839	0.3884	0.5498	0.6686	0.5356	0.4994
	Lawformer	**0.5765**†	**0.5647**†	**0.6203**†	**0.6480**†	**0.7641**†	**0.8091**†	**0.7404**†	**0.6638**†
	SAILER	0.5412	0.5471	0.5955	0.6124	0.7400	0.7719	0.7081	0.6347
	BERT-ts-L1	0.4941	0.5294	0.5745	0.6103	0.7505	0.7872	0.7160	0.6243
	Criminal-BERT	0.5059	0.5176	0.5664	0.5787	0.7465	0.7833	0.7028	0.6164
	SOUR	0.5176	0.5118	0.5775	0.5922	0.7403	0.7934	0.7086	0.6221
	ELCR	0.5294	0.5412	0.5734	0.5942	0.7610	0.7974	0.7175	0.6328
30%	condenser	0.3176	0.3647	0.4343	0.3070	0.4769	0.5891	0.4577	0.4149
	co-codenser	0.4235	0.4294	0.4849	0.3666	0.5285	0.6318	0.5090	0.4775
	Lawformer	0.4941	0.4765	0.5316	0.5058	0.6424	0.7218	0.6233	0.5620
	SAILER	0.5012	**0.5217**†	0.5650	0.5872	0.7375	0.7898	0.7048	0.6171
	BERT-ts-L1	0.4471	0.5118	0.5310	0.5697	0.7246	0.7693	0.6879	0.5923
	Criminal-BERT	0.5059	0.4588	0.5459	0.5461	0.7278	0.7621	0.6787	0.5911
	SOUR	0.4941	0.4529	0.5592	0.5457	0.7197	0.7791	0.6815	0.5918
	ELCR	**0.5294**†	0.5059	**0.5733**†	**0.6030**†	**0.7607**†	**0.7966**†	**0.7201**†	**0.6282**†
50%	condenser	0.3882	0.3647	0.4530	0.3182	0.4710	0.5942	0.4611	0.4316
	co-codenser	0.4235	0.4176	0.4853	0.3621	0.5200	0.6161	0.4994	0.4708
	Lawformer	0.3529	0.3176	0.4030	0.2809	0.4059	0.5100	0.3989	0.3784
	SAILER	0.3412	0.3765	0.4483	0.3348	0.4670	0.5903	0.4640	0.4264
	BERT-ts-L1	0.4000	0.4353	0.4876	0.4416	0.5418	0.6218	0.5351	0.4880
	Criminal-BERT	**0.5000**†	0.4529	0.5214	0.5358	0.6609	0.7131	0.6366	0.5640
	SOUR	0.4353	0.3882	0.4669	0.4043	0.5156	0.5902	0.5034	0.4668
	ELCR	0.4471	**0.4765**†	**0.5562**†	**0.5796**†	**0.7492**†	**0.7946**†	**0.7078**†	**0.6005**†
70%	condenser	0.2353	0.2941	0.3524	0.1870	0.2707	0.3660	0.2746	0.2843
	co-codenser	0.3059	0.3118	0.3742	0.2521	0.3279	0.3956	0.3252	0.3279
	Lawformer	0.2941	0.3176	0.3625	0.1735	0.2583	0.3355	0.2558	0.2903
	SAILER	0.3529	0.3235	0.4065	0.2857	0.3551	0.4348	0.3585	0.3598
	BERT-ts-L1	0.3294	0.2824	0.3706	0.2062	0.2897	0.3610	0.2856	0.3066
	Criminal-BERT	0.2235	0.2765	0.3645	0.1810	0.2655	0.3559	0.2675	0.2778
	SOUR	0.2235	0.2529	0.3522	0.1658	0.2379	0.3185	0.2407	0.2585
	ELCR	**0.3765**†	**0.3824**†	**0.4413**†	**0.3339**†	**0.4232**†	**0.5063**†	**0.4211**†	**0.4106**†
90%	condenser	0.2471	0.2824	0.3592	0.2139	0.2738	0.3539	0.2805	0.2884
	co-codenser	0.2471	0.2529	0.3388	0.1937	0.2572	0.3396	0.2635	0.2716
	Lawformer	0.1765	0.1706	0.2874	0.1243	0.1714	0.2312	0.1756	0.1936
	SAILER	0.1529	0.1882	0.3042	0.0965	0.1583	0.2177	0.1575	0.1863
	BERT-ts-L1	0.1294	0.1647	0.2736	0.0835	0.1300	0.1898	0.1344	0.1618
	Criminal-BERT	0.1529	0.1824	0.3025	0.1238	0.1641	0.2455	0.1778	0.1952
	SOUR	**0.4118**†	0.3471	**0.4508**†	**0.3708**†	0.3792	0.4121	0.3874	0.3953
	ELCR	0.3765	**0.3588**†	0.4430	0.3357	**0.4795**†	**0.5841**†	**0.4664**†	**0.4296**†

effectiveness of each component in ELCR. The model exhibits the greatest decline after the removal of noise detection and correction using evidential learning, which confirms the pivotal role of evidential learning in enhancing noise robustness.

5.4. Uncertainty analysis

To answer **RQ3**, we carry out experiments under 50% noisy correspondence on LeCaRD to visually investigate the evolution of evidence and uncertainty during the training process. Note that the evidence reflects the model's confidence in the relevance between a certain query-document pair, while the uncertainty refers to the model's uncertainty in matching a query to the corresponding document among all documents within the batch.

5.4.1. Uncertainty visualization

Fig. 3 and Fig. 4 respectively show the evolution of evidence and distinguishes four types of pair samples: clean positives, clean positives, noisy negatives (i.e., samples that are actually negative but labeled as positive due to noise), and noisy positives (i.e., samples that are actually positive but labeled as negative due to noise).

We observe a polarizing trend in the model's uncertainty and evidence value towards the training pair samples. We find the evidence values for noisy positives and clean positives gradually increase, while the evidence values for noisy negatives and cleans gradually decrease. Simultaneously, as the evidence value increases or decreases, the uncertainty undergoes a converse change. This

Table 5
Ablation study of ELCR on LeCaRD and ELAM with 50% noisy correspondence. The best performances are denoted in bold fonts. "w/o EL" stands for the variant where the evidential learning-based uncertainty estimation is not employed. "w/o MFR" denotes the variant where multi-faceted relevance modeling is not utilized. "w/o LC" represents the variant without label correction. Paired t-tests are conducted and "†" indicates the best performances are statistically significant compared with the second-best (p-value < 0.05).

Dataset	Setup	NDCG@10	NDCG@20	NDCG@30
LeCaRD	ELCR	**0.6662**†	**0.7019**†	**0.7354**†
	w/o EL	0.6619	0.6913	0.7062
	w/o MFR	0.6580	0.6810	0.7239
	w/o LC	0.6638	0.6886	0.7218
	Best baseline	0.5562	0.5839	0.5993
ELAM	ELCR	**0.5796**†	**0.7492**†	**0.7946**†
	w/o EL	0.5709	0.7116	0.7689
	w/o MFR	0.5735	0.7416	0.7823
	w/o LC	0.5857	0.7460	0.7864
	Best baseline	0.5358	0.6609	0.7131

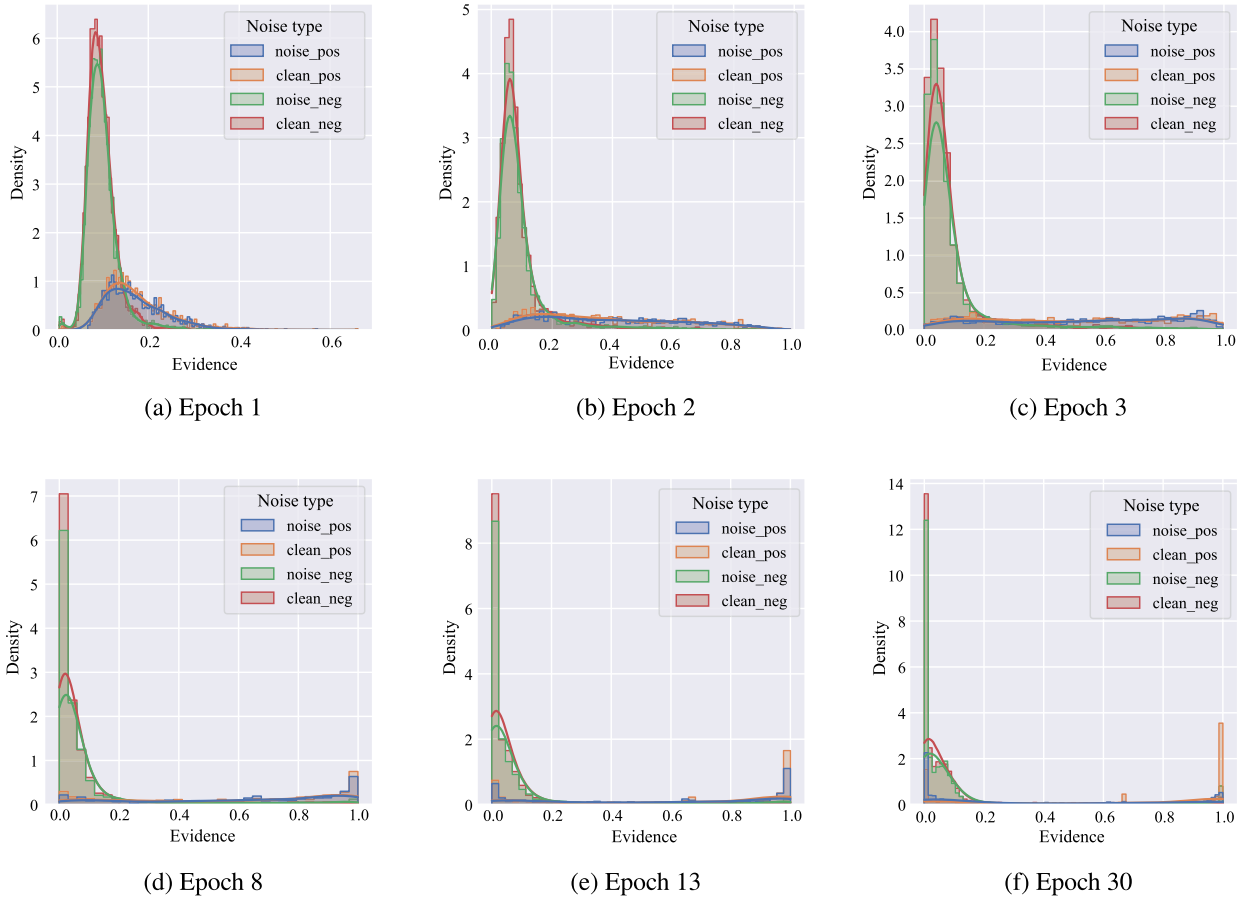


Fig. 3. Visualization of the evidence distribution of clean positive pairs, noisy positive pairs, clean negative pairs, and noisy negative pairs at different training epochs of ELCR on LeCaRD with 50% noisy correspondence. The horizontal axis represents a certain evidence value, and the corresponding vertical axis value represents the relative density of samples at that corresponding evidence value. The distribution changes of evidence indicate that the model gradually becomes capable of distinguishing between positive and negative samples during training.

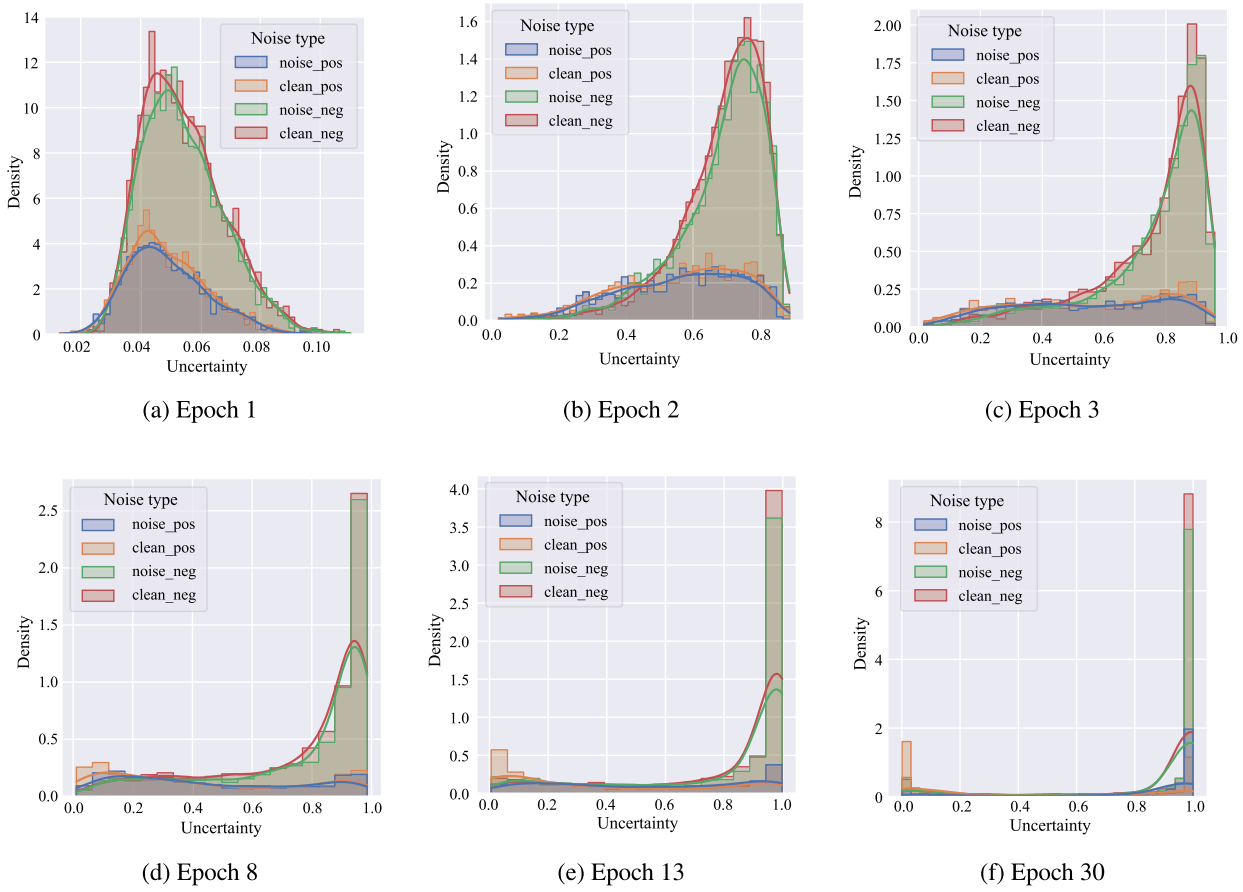


Fig. 4. Visualization of the uncertainty distribution of clean positive pairs, noisy positive pairs, clean negative pairs, and noisy negative pairs at different training epochs of ELCR on LeCaRD with 50% noisy correspondence. The horizontal axis represents a certain uncertainty value, and the corresponding vertical axis value represents the relative density of samples at that corresponding uncertainty value. The distribution changes of uncertainty indicate that the model gradually becomes capable of distinguishing between positive and negative samples during training.

indicates that the evidence can progressively distinguish between actual positive and negative samples. In addition, compared to noisy samples, the distribution of clean samples becomes increasingly concentrated around 0 or 1, verifying the model's learned ability to discriminate against noise. Besides, we also find the model can filter out clean positive samples through evidence values, but it cannot provide sufficient evidence for noisy positive samples. Even though a legal pre-trained model only requires a small number of clean positive samples for fine-tuning to achieve good performance, the inability to differentiate noisy positive samples may still lead to a reduced number of available positive samples, resulting in a possible decrease in model performance, especially under conditions of a large amount of noise.

5.4.2. Uncertainty quantification

To answer **RQ3** in a quantitative way, we select 50% noisy correspondence for ELCR training, and collect the average evidence and uncertainty values before and after optimization for both ELAM and LeCaRD.[3] In our noisy correspondence scenario, only samples with clean and relevant labels contribute positive signals for model training. In other words, a well-trained model should assign higher evidence values and lower uncertainty scores to these clean and relevant samples compared to polluted or irrelevant ones.

Accordingly, we recorded the evidence values and uncertainty scores for the clean, relevant samples before and after model training. As shown in Table 6, we observe the following trends across both LeCaRD and ELAM datasets: (1) evidence values for clean, relevant samples increase significantly after training, while they decrease notably for other pairs; (2) uncertainty scores for clean, relevant pairs decrease significantly after optimization, while they increase for other pairs.

These results clearly demonstrate the effectiveness of ELCR in identifying noisy correspondences through evidence and uncertainty metrics. On the one hand, the increase in evidence and decrease in uncertainty for clean, relevant pairs suggest that the model is progressively recognizing the true relevance signals. On the other hand, the decrease in evidence and increase in uncertainty for other pairs further indicate that the model's ability to differentiate between relevant and noisy pairs is improving, enhancing its robustness

[3] In this section, both evidence and uncertainty have been normalized to facilitate comparison, respectively.

Table 6
Comparison of evidence and uncertainty for clean relevant and other sample pairs before and after ELCR optimization on LeCaRD and ELAM with 50% noisy correspondence. Evidence and uncertainty have both been normalized for easy comparison. The model's increased evidence and decreased uncertainty for clean relevant pairs show its growing ability to recognize the clean relevance between these samples. Conversely, the decreased evidence and increased uncertainty for other pairs highlight the model's enhanced distinguishing capability and improved robustness against noise.

Dataset		LeCaRD		ELAM	
Sample Pairs		Clean Relevant	Others	Clean Relevant	Others
Evidence	Original	0.3806	0.1418	0.5044	0.1310
	Optimized	0.6205	0.1284	0.5695	0.0939
Change (%)		+63.04%	−9.50%	+12.90%	−28.31%
Uncertainty	Original	0.5611	0.6700	0.2139	0.3207
	Optimized	0.4133	0.8309	0.1842	0.4324
Change (%)		−26.33%	+24.03%	−13.88%	+34.83%

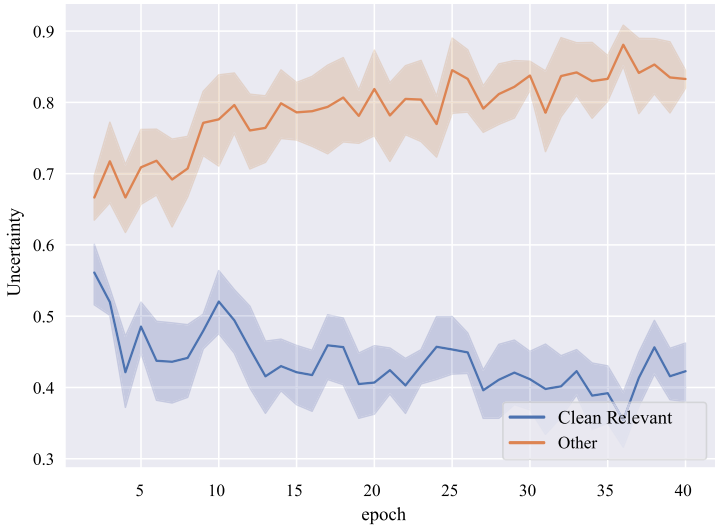


Fig. 5. The trend of uncertainty for clean relevant samples (in blue) and others (in orange) on the LeCaRD dataset across training epochs, with 95% confidence intervals based on five repeated experiments. This trend demonstrates ELCR's progressive recognition of true relevance signals through uncertainty modeling.

against noise. In addition, we observe that even before optimization, the model assigns higher uncertainty to noisy pairs compared to clean, relevant pairs. This suggests that the inherent knowledge of the pre-trained legal model can, to some extent, distinguish noisy legal text pairs, as such noisy associations tend to contradict the legal common sense embedded in the pre-trained data.

Furthermore, we also illustrate the distinction of uncertainty in the LeCaRD dataset at a 95% confidence interval in Fig. 5. As the number of training epochs increases, the model's ability to distinguish between samples gradually improves, leading to a clearer differentiation between these two different pairs.

5.5. Cross-model adaptability

As described above, ELCR is a model-agnostic framework. To answer **RQ4**, we incorporate various backbone models into ELCR and explore the cross-model adaptability of ELCR.

Specifically, we select two popular models pre-trained on legal data (Criminal-BERT [40] and BERT-ts-L1 [36]) as well as a language model designed for retrieval tasks (Condenser [43]) as the backbone. We conduct comparative experiments on three backbones and two datasets. Specifically, we first assess the performance of the three models in their original configurations, and then integrate ELCR into these models and evaluate their performance.

Table 7 illustrates the cross-model adaptation performance of ELCR method on different backbones. We find the application of ELCR yields significant performance enhancements across models on both datasets. For the Condenser model, the average improvement is approximately 7.2%. The BERT-ts-L1 model experienced a more substantial average improvement of 15.8%. The Criminal-BERT model demonstrates the most remarkable average improvement, with a figure of 24.9%. These results underscore the efficacy of the ELCR method in enhancing cross-model adaptability. It is noteworthy that the improvement observed on Condenser

Table 7
Cross-model adaptation performance of ELCR method on different backbones. The table shows the comparison of model performance before and after applying the ELCR method on two datasets (ELAM and LeCaRD) and three models (Condenser, BERT-ts-L1, Criminal-BERT). "Vanilla" refers to the original version of the corresponding backbone without applying ELCR. "Improvement" refers to the mean percentage increase in performance metrics (NDCG@10, NDCG@20, NDCG@30) after the application of the ELCR method compared to the "Vanilla" version of the models.

Dataset	Backbone	Setup	NDCG@10	NDCG@20	NDCG@30	Improvement (%)
LeCaRD	Condenser	Vanilla	0.3118	0.3463	0.3943	4.6
		ELCR	0.3329	0.3578	0.3988	
	BERT-ts-L1	Vanilla	0.5562	0.5539	0.5793	20.7
		ELCR	0.6662	0.7019	0.7354	
	Criminal-BERT	Vanilla	0.5395	0.5559	0.5765	23.8
		ELCR	0.6583	0.7036	0.7397	
ELAM	Condenser	Vanilla	0.3182	0.4710	0.5942	9.7
		ELCR	0.3565	0.4946	0.6259	
	BERT-ts-L1	Vanilla	0.5358	0.6609	0.7131	10.9
		ELCR	0.5796	0.7492	0.7946	
	Criminal-BERT	Vanilla	0.4416	0.5418	0.6218	26.1
		ELCR	0.5638	0.7375	0.7658	

Table 8
Retrieval performance comparison of ELCR and uncertainty-based methods on LeCaRD and ELAM with 50% noisy correspondences. The best performances are highlighted in bold.

Dataset	Method	NDCG@10	NDCG@20	NDCG@30
LeCaRD	DBAESS	0.5681	0.6122	0.6362
	ERDL	0.5526	0.5938	0.6218
	IEP	0.5221	0.5679	0.5995
	ESC	0.5852	0.6353	0.6791
	ELCR	**0.6662**	**0.7019**	**0.7354**
ELAM	DBAESS	0.5491	0.6763	0.7208
	ERDL	0.5116	0.6397	0.6743
	IEP	0.4916	0.5318	0.5918
	ESC	0.5301	0.6518	0.6987
	ELCR	**0.5796**	**0.7492**	**0.7946**

is less pronounced compared to the other two models. This indicates that models pre-trained on legal data are more compatible with ELCR.

In short, these results verify the cross-model adaptability of ELCR, showcasing its potential as a robust choice for legal case retrieval with noisy correspondences.

5.6. Comparison with uncertainty-based denoising methods

To further evaluate the robustness of ELCR against noisy correspondences, we select 50% noisy correspondence and compare ELCR with several state-of-the-art uncertainty-based denoising methods, including DBAESS [18], ERDL [34], IEP [18] and ESC [16]. Given the differences in data types and task setups, we adapt these methods to fit the legal case retrieval scenario, as described in Sec. 5.1.2. Table 8 presents the comparative results. From these results, we observe that ELCR consistently outperforms the baselines in terms of NDCG, demonstrating its superior noise robustness in the legal case retrieval task. While the baseline methods exhibit some effectiveness, they are less robust compared to ELCR. We attribute ELCR's superior performance to its explicit modeling of legal uncertainty, its multi-faceted understanding of similarity in legal cases, and its effective noise identification and correction process tailored to the relevance ranking task.

6. Conclusion

This paper proposes a novel evidential learning framework called ELCR to estimate legal uncertainty and enhance robustness against noisy correspondences in legal case retrieval. ELCR predicts legal relevance between queries and candidates from the perspectives of concept, rule, and fact. This multi-faceted relevance is used to estimate evidence-based uncertainty under the Dempster-Shafer Evidence Theory. Explicit uncertainty estimation aids in detecting and correcting observed labels. Additionally, we introduce two evidence-based training objectives to further improve the robustness of ELCR. In experiments conducted on two benchmark datasets,

LeCaRD and ELAM, our ELCR achieves an average NDCG improvement of 20.90% and 10.99%, respectively, compared to various baselines, including learning-to-rank methods and uncertainty-based denoising methods. These results demonstrate that our method is robust against noisy correspondences while maintaining competitive retrieval performance.

CRediT authorship contribution statement

Weicong Qin: Writing – original draft, Validation, Software, Methodology, Investigation. **Weijie Yu:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Kepu Zhang:** Writing – review & editing, Validation, Methodology, Investigation. **Haiyuan Zhao:** Writing – original draft, Validation, Methodology, Investigation. **Jun Xu:** Supervision, Project administration, Funding acquisition. **Ji-Rong Wen:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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